

$$11) \tan \theta = \frac{1}{2}$$

$$\text{To prove } \frac{1}{2} \sin \theta \cdot \cos \theta = \frac{1}{5}$$

$$\frac{\sin \theta}{\cos \theta} = \frac{1}{2} \quad \left| \quad \sin \theta = \frac{1}{2} \cos^2 \theta \right.$$

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$$\cos \theta = 2$$

$$\frac{\sin \theta}{\cos \theta} = \frac{1}{2}$$

$$\frac{1 \times \cos \theta}{2 \sin \theta}$$

$$\frac{\sin \theta \cos \theta}{\cos^2 \theta} = \frac{1 \cos \theta}{2}$$

$$\frac{\sin \theta \cdot \cos \theta}{\cos \theta}$$

$$2 \times 2 \times 2 \times 2 \times 2$$

$$2 \times 2 \times 2$$

$$2^4 \times 8$$

$$5000$$

$$5 \times 5 \times 5 \times 5$$

$$\frac{125 \times 5}{25}$$

$$12) \frac{257}{5000} = \frac{257}{2^3 \times 5^4}$$

12)

13) Let us assume  $\sqrt{3}$  is a rational number.  
i.e.  $\sqrt{3} = \frac{a}{b}$  [where  $a$  &  $b$  are co-primes &  $b \neq 0$ ]

$$\sqrt{3} \cdot b = a \quad [\text{squaring both sides}]$$

$$3b^2 = a^2$$

3 divides  $a^2 \rightarrow$  3 divides  $a$  also

$$\text{i.e. } a = 3c$$

$$3b^2 = (3c)^2$$

$$3b^2 = 9c^2$$

$$b^2 = 3c^2$$

$\therefore b^2$  has more than one factors.  
i.e. Our assumption is wrong.  
 $\sqrt{3}$  is an irrational number.

$$\left(\frac{1}{\sqrt{5}}\right)^2 = \frac{1}{2} \sin \theta \cdot \cos \theta$$

$$\frac{1}{2} \times \sin \theta \cdot \cos \theta = \frac{1}{5}$$

Hence, proved

$\sin \theta = \frac{\text{Opp.}}{\text{hypotenuse}}$

$$= \frac{1}{\sqrt{5}}$$